

Phase 4 : Queries Report

SCHEMA:

The schema is displayed in EER_SCHEMA.png. It has also been summarized below.

coordinated_by(employee_id, program_id)

participates_in(program_id, employee_id, end_date, program_name, start_date)

health_metrics(employee_id, date_measured, cholesterol_levels, resting_heart_rate,
blood_pressure_systolic, blood_pressure_diastolic, bmi)

wellness_coordinator(employee_id, area_of_expertise, coordinator_credentials)

wellness_program(program_id, employee_id, end_date, program_name, start_date, type)

employee(employee_id, fname, middle_initial, lname, role, phone_number, department_id,
work_email)

Table Creation Commands:

The “init.build.sql” script can build the tables. Below are the table creation commands listed in the build script.

```
CREATE TABLE employee (  
    employee_id INT PRIMARY KEY,  
    fname VARCHAR(20),  
    middle_initial CHAR(1),  
    lname VARCHAR(50),  
    `role` ENUM('worker', 'secretary', 'coordinator'),  
    phone_number VARCHAR(20),  
    department_id ENUM('sales', 'marketing', 'finance', 'operations'), -- (sales, marketing, finance,  
operations)  
    work_email VARCHAR(30)  
);
```

```
CREATE TABLE wellness_coordinator (  
    employee_id INT PRIMARY KEY,  
    area_of_expertise VARCHAR(20),  
    coordinator_credentials VARCHAR(10),  
    FOREIGN KEY (employee_id) REFERENCES employee(employee_id)  
        ON DELETE CASCADE  
        ON UPDATE CASCADE  
);
```

```
CREATE TABLE wellness_program (  
    program_id INT PRIMARY KEY,  
    employee_id INT,  
    end_date DATE,  
    program_name VARCHAR(20),  
    start_date DATE,  
    `type` ENUM('bmi', 'blood_pressure', 'heart_rate', 'cholesterol'),  
    FOREIGN KEY (employee_id) REFERENCES wellness_coordinator(employee_id)  
        ON DELETE CASCADE  
        ON UPDATE CASCADE  
);
```

```
CREATE TABLE participates_in (  
    employee_id INT,  
    program_id INT,  
    enrollment_date DATE,  
    PRIMARY KEY (employee_id, program_id),  
    FOREIGN KEY (employee_id) REFERENCES employee(employee_id)  
        ON DELETE CASCADE
```

```

        ON UPDATE CASCADE,
FOREIGN KEY (program_id) REFERENCES wellness_program(program_id)
        ON DELETE CASCADE
        ON UPDATE CASCADE
);

```

```

CREATE TABLE health_metrics (
    employee_id INT,
    date_measured DATE,
    cholesterol_levels INT,
    resting_heart_rate INT,
    blood_pressure_systolic INT,
    blood_pressure_diastolic INT,
    bmi FLOAT,
    PRIMARY KEY (employee_id, date_measured),
    FOREIGN KEY (employee_id) REFERENCES employee(employee_id)
        ON DELETE CASCADE
        ON UPDATE CASCADE
);

```

```

CREATE TABLE coordinated_by (
    employee_id INT,
    program_id INT,
    PRIMARY KEY (employee_id, program_id),
    FOREIGN KEY (employee_id) REFERENCES wellness_coordinator(employee_id)
        ON DELETE NO ACTION
        ON UPDATE CASCADE,
    FOREIGN KEY (program_id) REFERENCES wellness_program(program_id)
        ON DELETE CASCADE -- We will not allow for a coordinator to be deleted from the database.
        ON UPDATE CASCADE -- It would delete all programs they manage. Updating their ID should
be fine
);

```

Table Contents:

Below are the contents of the tables, displayed using “SELECT * {table_name}”

coordinated_by

```
mysql> SELECT * FROM coordinated_by;
```

employee_id	program_id
103	201
113	201
105	202
107	203
121	204
122	205
123	206
124	207
125	208
126	209
126	210

employee

The contents of this table are available for viewing in excel, via “employee_contents.csv”

```
mysql> SELECT * FROM employee;
```

employee_id	fname	middle_initial	lname	role	phone_number	department_id	work_email
101	John	A	Smith	worker	555-0101	finance	john.smith@gre.com
102	Emma	B	Johnson	worker	555-0102	marketing	emma.johnson@gre.com
103	Michael	C	Williams	coordinator	555-0103	operations	michael.williams@gre.com
104	Olivia	D	Brown	worker	555-0104	sales	olivia.brown@gre.com
105	William	E	Jones	coordinator	555-0105	operations	william.jones@gre.com
106	Sophia	F	Garcia	worker	555-0106	marketing	sophia.garcia@gre.com
107	James	G	Miller	coordinator	555-0107	operations	james.miller@gre.com
108	Isabella	H	Davis	worker	555-0108	sales	isabella.davis@gre.com
109	Benjamin	I	Rodriguez	worker	555-0109	marketing	benjamin.rodriguez@gre.com
110	Mia	J	Martinez	worker	555-0110	finance	mia.martinez@gre.com
111	Jacob	K	Hernandez	worker	555-0111	sales	jacob.hernandez@gre.com
112	Charlotte	L	Lopez	worker	555-0112	marketing	charlotte.lopez@gre.com
113	Ethan	M	Gonzalez	coordinator	555-0113	operations	ethan.gonzalez@gre.com
114	Amelia	N	Wilson	worker	555-0114	sales	amelia.wilson@gre.com
115	Alexander	O	Anderson	worker	555-0115	finance	alexander.anderson@gre.com
116	Harper	P	Thomas	worker	555-0116	sales	harper.thomas@gre.com
117	Daniel	Q	Taylor	secretary	555-0117	operations	daniel.taylor@gre.com
118	Evelyn	R	Moore	secretary	555-0118	finance	evelyn.moore@gre.com
119	Matthew	S	Jackson	secretary	555-0119	marketing	matthew.jackson@gre.com
120	Abigail	T	Martin	secretary	555-0120	sales	abigail.martin@gre.com
121	Adam	R	Samuel	coordinator	555-1067	operations	adam.samuel@gre.com
122	Rachael	E	Johnson	coordinator	555-0123	operations	rachael.johnson@gre.com
123	Mary	A	Grace	coordinator	555-0159	operations	mary.grace@gre.com
124	William	D	Oracle	coordinator	555-8924	operations	william.oracle@gre.com

125	Patrick	A	Star	coordinator	555-1756	operations	patrick.star@gre.com
126	Bob	T	Sponge	coordinator	555-6928	operations	bob.sponge@gre.com

health_metrics

```
mysql> SELECT * FROM health_metrics;
```

employee_id	date_measured	cholesterol_levels	resting_heart_rate	blood_pressure_systolic	blood_pressure_diastolic	bmi
101	2023-09-01	180	72	120	80	24.5
101	2023-10-01	178	71	119	79	24.3
101	2023-11-01	176	70	118	78	24.1
102	2023-09-01	190	75	122	82	23
102	2023-10-01	189	74	121	81	22.9
102	2023-11-01	188	73	120	80	22.8
103	2023-09-01	170	70	118	78	26.1
103	2023-10-01	169	69	117	77	26
103	2023-11-01	168	68	116	76	25.9
104	2023-09-01	160	68	115	75	22.5
104	2023-10-01	160	68	115	75	22.5
104	2023-11-01	160	68	115	75	22.5
105	2023-09-01	200	78	125	85	28
105	2023-10-01	198	76	123	83	27.8
105	2023-11-01	196	74	121	81	27.6
106	2023-09-01	193	78	119	80	27.7
106	2023-10-01	192	75	118	79	27.4
106	2023-11-01	189	71	112	80	27.9
107	2023-09-01	178	97	143	90	39
107	2023-10-01	178	94	141	91	38.6
107	2023-11-01	178	89	138	88	38.1
108	2023-09-01	153	60	119	82	20.1
108	2023-10-01	154	61	119	81	19.8
108	2023-11-01	152	58	119	83	19.9
109	2023-09-01	250	74	130	85	29.5
109	2023-10-01	248	73	129	84	29.3
109	2023-11-01	246	72	128	83	29
110	2023-09-01	240	76	127	82	28.1
110	2023-10-01	238	75	126	81	27.8
110	2023-11-01	236	74	125	80	27.5
111	2023-09-01	260	82	135	88	30
111	2023-10-01	258	80	134	87	29.8
111	2023-11-01	256	78	133	86	29.5
112	2023-09-01	245	70	122	81	26
112	2023-10-01	243	69	121	80	25.8
112	2023-11-01	241	68	120	79	25.5

participates_in

```
mysql> SELECT * FROM participates_in;
```

employee_id	program_id	enrollment_date
101	201	2023-09-05
102	201	2023-09-06
103	201	2023-09-07
104	201	2023-09-08
105	202	2023-09-16
106	202	2023-09-17
107	202	2023-09-18
108	202	2023-09-19
109	203	2023-10-01
110	203	2023-10-02
111	203	2023-10-03
112	203	2023-10-04

wellness_coordinator

```
mysql> SELECT * FROM wellness_coordinator;
```

employee_id	area_of_expertise	coordinator_credentials
103	bmi	CERTFIT
105	blood_pressure	CERTMIN
107	heart_rate	CERTNUT
113	cholesterol	CERTGEN
121	bmi	CERTPOP
122	blood_pressure	CERTPOW
123	heart_rate	CERTNAB
124	cholesterol	CERTCON
125	bmi	CERTGRE
126	blood_pressure	CERTIFY

wellness_program

```
mysql> SELECT * FROM wellness_program;
```

program_id	employee_id	end_date	program_name	start_date	type
201	103	2026-12-31	Fitness Challenge	2023-09-01	bmi
202	105	2026-12-31	Mindfulness Workshop	2023-09-15	heart_rate
203	107	2026-12-31	Nutrition Plan	2023-09-30	cholesterol
204	113	2025-12-31	Clean Eating	2024-01-01	blood_pressure
205	121	2024-06-30	Take the Stairs	2024-01-15	bmi
206	122	2024-06-30	Deep Breathing	2024-02-01	heart_rate
207	123	2024-06-30	No Sugar Challenge	2024-03-01	cholesterol
208	124	2024-06-30	Sleep and Sobriety	2024-03-15	blood_pressure
209	125	2024-06-30	No Sugar	2024-04-01	bmi
210	126	2024-06-30	Daily Movement	2024-04-15	heart_rate

Queries:

Query 1:

Description: Query 1 is designed to find the number of participants in program 203 grouped by department.

Query:

```
SELECT department_id, count(*)
FROM employee, participates_in, wellness_program
WHERE participates_in.program_id = wellness_program.program_id
      AND wellness_program.program_id = 203
      AND participates_in.employee_id = employee.employee_id
GROUP BY department_id;
```

Results:

department_id	count(*)
marketing	2
finance	1
sales	1

Query 2:

Description: Query 2 lists all the wellness programs by program ID, and also displays the amount of participants that are enrolled in each program.

Query:

```
SELECT wellness_program.program_id, count(*)
FROM wellness_program, participates_in, employee
WHERE
    wellness_program.program_id = participates_in.program_id
AND participates_in.employee_id = employee.employee_id
GROUP BY program_id;
```

Results:

program_id	count(*)
201	4
202	4
203	4

Query 3:

Description: Query 3 displays the average systolic and diastolic blood pressure of the employees for each department

Query:

```
SELECT e.department_id,  
       ROUND(AVG(h.blood_pressure_systolic), 2) AS avg_systolic,  
       ROUND(AVG(h.blood_pressure_diastolic), 2) AS avg_diastolic  
FROM employee e  
JOIN health_metrics h  
ON e.employee_id = h.employee_id  
GROUP BY e.department_id;
```

Results:

department_id	avg_systolic	avg_diastolic
finance	122.50	80.00
marketing	121.83	81.17
operations	126.89	83.22
sales	122.67	81.33

Query 4:

Description: This query returns each programs program ID as well as the contact information of the programs coordinator

Query:

```
SELECT distinct fname, lname, work_email, phone_number, coordinated_by.program_id
FROM(
SELECT *
FROM employee
WHERE `role` = 'coordinator'
) AS contacts,
wellness_program, coordinated_by
WHERE
wellness_program.program_id = coordinated_by.program_id
AND coordinated_by.employee_id = contacts.employee_id;
```

Results:

fname	lname	work_email	phone_number	program_id
Michael	Williams	michael.williams@gre.com	555-0103	201
Ethan	Gonzalez	ethan.gonzalez@gre.com	555-0113	201
William	Jones	william.jones@gre.com	555-0105	202
James	Miller	james.miller@gre.com	555-0107	203
Adam	Samuel	adam.samuel@gre.com	555-1067	204
Rachael	Johnson	rachael.johnson@gre.com	555-0123	205
Mary	Grace	mary.grace@gre.com	555-0159	206
William	Oracle	william.oracle@gre.com	555-8924	207
Patrick	Star	patrick.star@gre.com	555-1756	208
Bob	Sponge	bob.sponge@gre.com	555-6928	209
Bob	Sponge	bob.sponge@gre.com	555-6928	210

Query 5:

Description: This query returns the names as well as the latest health metrics for all employees.

Query:

```
SELECT fname as 'First Name',
       lname as 'Last Name',
       resting_heart_rate as 'Resting BPM',
       cholesterol_levels as 'Cholesterol',
       blood_pressure_systolic as 'Sys',
       blood_pressure_diastolic as 'DBP',
       bmi as 'BMI'
FROM employee
LEFT JOIN health_metrics
ON employee.employee_id = health_metrics.employee_id
WHERE date_measured = (
    (SELECT MAX(date_measured)
     FROM health_metrics, employee
     WHERE health_metrics.employee_id = employee.employee_id)
);
```

Results:

First Name	Last Name	Resting BPM	Cholesterol	Sys	DBP	BMI
John	Smith	70	176	118	78	24.1
Emma	Johnson	73	188	120	80	22.8
Michael	Williams	68	168	116	76	25.9
Olivia	Brown	68	160	115	75	22.5
William	Jones	74	196	121	81	27.6
Sophia	Garcia	71	189	112	80	27.9
James	Miller	89	178	138	88	38.1
Isabella	Davis	58	152	119	83	19.9
Benjamin	Rodriguez	72	246	128	83	29
Mia	Martinez	74	236	125	80	27.5
Jacob	Hernandez	78	256	133	86	29.5
Charlotte	Lopez	68	241	120	79	25.5

Query 6:

Description: Query 6 finds employees who are not participating in any wellness program. Their names as well as emails are displayed.

Query:

```
SELECT fname, lname, work_email
FROM employee
WHERE employee_id
NOT IN (
SELECT employee_id
FROM participates_in
);
```

Results:

fname	lname	work_email
Ethan	Gonzalez	ethan.gonzalez@gre.com
Amelia	Wilson	amelia.wilson@gre.com
Alexander	Anderson	alexander.anderson@gre.com
Harper	Thomas	harper.thomas@gre.com
Daniel	Taylor	daniel.taylor@gre.com
Evelyn	Moore	evelyn.moore@gre.com
Matthew	Jackson	matthew.jackson@gre.com
Abigail	Martin	abigail.martin@gre.com
Adam	Samuel	adam.samuel@gre.com
Rachael	Johnson	rachael.johnson@gre.com
Mary	Grace	mary.grace@gre.com
William	Oracle	william.oracle@gre.com
Patrick	Star	patrick.star@gre.com
Bob	Sponge	bob.sponge@gre.com

Query 7:

Description: Query 7 identifies employees who have shown improvement in health metrics between the two most recent measurements

Query:

```
SELECT h1.employee_id, e.fname, e.lname,  
h1.bmi AS Latest_BMI, h2.bmi AS Previous_BMI,  
h1.resting_heart_rate AS Latest_Heart_Rate, h2.resting_heart_rate AS Previous_Heart_Rate  
FROM health_metrics h1  
JOIN health_metrics h2 ON h1.employee_id = h2.employee_id  
JOIN employee e ON h1.employee_id = e.employee_id  
WHERE h1.date_measured > h2.date_measured  
AND h1.date_measured = (SELECT MAX(date_measured) FROM health_metrics WHERE  
employee_id = h1.employee_id)  
AND h2.date_measured = (SELECT MAX(date_measured) FROM health_metrics WHERE  
employee_id = h2.employee_id AND date_measured < h1.date_measured)  
AND (h1.bmi < h2.bmi OR h1.resting_heart_rate < h2.resting_heart_rate);
```

Results:

employee_id	fname	lname	Latest_BMI	Previous_BMI	Latest_Heart_Rate	Previous_Heart_Rate
101	John	Smith	24.1	24.3	70	71
102	Emma	Johnson	22.8	22.9	73	74
103	Michael	Williams	25.9	26	68	69
105	William	Jones	27.6	27.8	74	76
106	Sophia	Garcia	27.9	27.4	71	75
107	James	Miller	38.1	38.6	89	94
108	Isabella	Davis	19.9	19.8	58	61
109	Benjamin	Rodriguez	29	29.3	72	73
110	Mia	Martinez	27.5	27.8	74	75
111	Jacob	Hernandez	29.5	29.8	78	80
112	Charlotte	Lopez	25.5	25.8	68	69

Query 8:

Description: Query 8 identifies employees whose BMI, cholesterol levels, or blood pressure are very poor, and returns their name and all bmi, cholesterol, and blood pressure information alongside the date of their metric snapshot.

All health snapshots with concerning health metrics are returned.

Query:

```
SELECT distinct e.fname, e.lname, h.bmi, h.cholesterol_levels, h.blood_pressure_systolic,
h.blood_pressure_diastolic, h.date_measured
FROM employee e
JOIN health_metrics h ON e.employee_id = h.employee_id
WHERE (h.bmi < 18.5 OR h.bmi > 24.9)
OR (h.cholesterol_levels > 200)
OR (h.blood_pressure_systolic > 130 OR h.blood_pressure_diastolic > 90);
```

Results:

fname	lname	bmi	cholesterol_levels	blood_pressure_systolic	blood_pressure_diastolic	date_measured
Michael	Williams	26.1	170	118	78	2023-09-01
Michael	Williams	26	169	117	77	2023-10-01
Michael	Williams	25.9	168	116	76	2023-11-01
William	Jones	28	200	125	85	2023-09-01
William	Jones	27.8	198	123	83	2023-10-01
William	Jones	27.6	196	121	81	2023-11-01
Sophia	Garcia	27.7	193	119	80	2023-09-01
Sophia	Garcia	27.4	192	118	79	2023-10-01
Sophia	Garcia	27.9	189	112	80	2023-11-01
James	Miller	39	178	143	90	2023-09-01
James	Miller	38.6	178	141	91	2023-10-01
James	Miller	38.1	178	138	88	2023-11-01
Benjamin	Rodriguez	29.5	250	130	85	2023-09-01
Benjamin	Rodriguez	29.3	248	129	84	2023-10-01
Benjamin	Rodriguez	29	246	128	83	2023-11-01
Mia	Martinez	28.1	240	127	82	2023-09-01
Mia	Martinez	27.8	238	126	81	2023-10-01
Mia	Martinez	27.5	236	125	80	2023-11-01
Jacob	Hernandez	30	260	135	88	2023-09-01
Jacob	Hernandez	29.8	258	134	87	2023-10-01
Jacob	Hernandez	29.5	256	133	86	2023-11-01
Charlotte	Lopez	26	245	122	81	2023-09-01
Charlotte	Lopez	25.8	243	121	80	2023-10-01
Charlotte	Lopez	25.5	241	120	79	2023-11-01